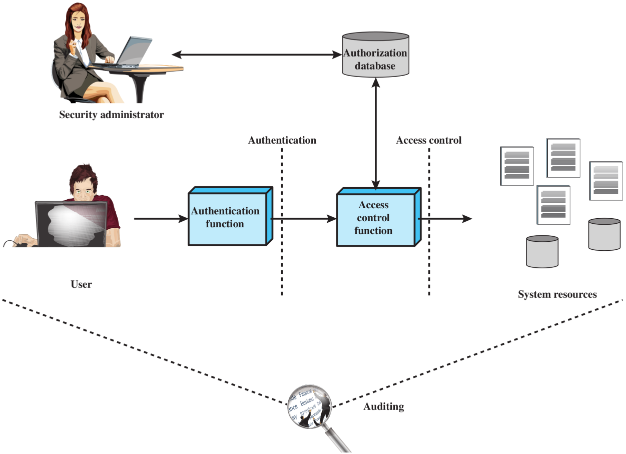
Access Control

Contents

The prevention of unauthorized use of a resource, including the prevention of use of a resource in an unauthorized manner.

— ITU-T Recommendation X.800 "Security architecture for Open Systems Interconnection"

Relationship Among Access Control and Other Security Functions



Credit: Figure 4.1 in Stallings and Brown, *Computer Security*, 2nd Ed., Pearson 2012

Access Control and Other Security Functions

Authentication
: verification that the credentials of a user or other entity are valid

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Audit independent review of system records and activities in order to test for adequacy of system control, ensure compliance to policy, detect breaches and recommend changes

Access Control Policies

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Role-based Access Control roles of users in system and rules for roles are used to control access

DAC, MAC and RBAC are not mutually exclusive

General Requirements of Access Control

- Reliable input
- Fine and coarse specifications
- Least privilege
- Separation of duty
- Open and closed policies
- Policy combinations and conflict resolution
- Administrative policies
- Dual control

Basic Elements of Access Control System

Subject entity capable of access resources

- Often subject is a software process
- Classes of subject, e.g. Owner, Group, World

Object resource to which access is controlled

- E.g. records, blocks, pages, files, portions of files, directories, email boxes, programs, communication ports

Access right describes way in which a subject may access an object

- E.g. read, write, execute, delete, create, search

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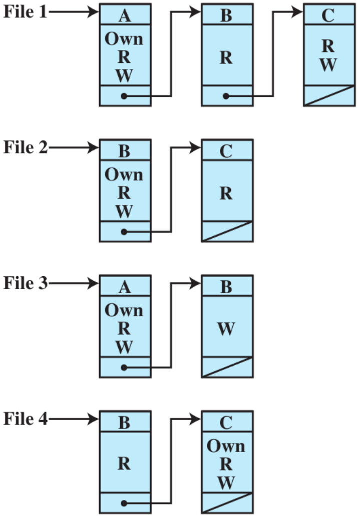
  Capability Lists For each subject, list objects and the rights the subject have on that object
- Alternative implementation: authorization table listing subject, access mode and object; easily implemented in database

Example of DAC Access Matrix

		OBJECTS			
		File 1	File 2	File 3	File 4
SUBJECTS	User A	Own Read Write		Own Read Write	
	User B	Read	Own Read Write	Write	Read
	User C	Read Write	Read		Own Read Write

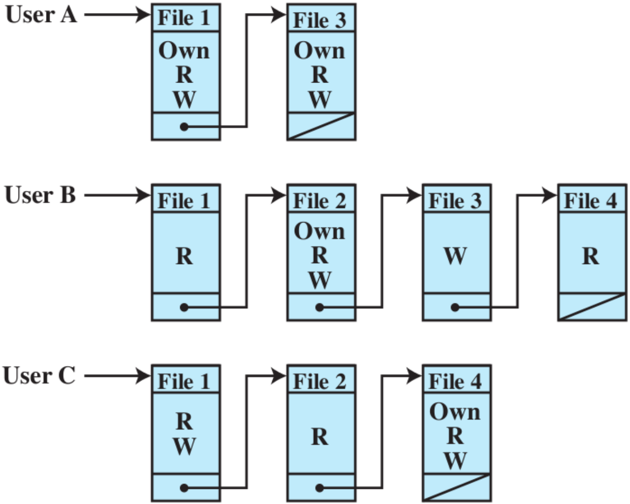
Credit: Figure 4.3(a) in Stallings and Brown, *Computer Security*, 2nd Ed., Pearson 2012

Example of Access Control Lists



Credit: Figure 4.3(b) in Stallings and Brown, *Computer Security*, 2nd Ed., Pearson 2012

Example of Capability Lists



Credit: Figure 4.3(c) in Stallings and Brown, *Computer Security*, 2nd Ed., Pearson 2012

Example of Authorization Table

Subject	Access Mode	Object
A	Own	File 1
A	Read	File 1
A	Write	File 1
A	Own	File 3
A	Read	File 3
A	Write	File 3
B	Read	File 1
B	Own	File 2
B	Read	File 2
B	Write	File 2
B	Write	File 3
B	Read	File 4
C	Read	File 1
C	Write	File 1
C	Read	File 2
C	Own	File 4
C	Read	File 4
C	Write	File 4

Credit: Table 4.1 in Stallings and Brown, *Computer Security*, 2nd Ed., Pearson 2012

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- Users may be assigned multiple roles; static or dynamic
- Sessions are temporary assignments of user to role(s)
- Access control matrix can map users to roles and roles to objects

	R_1	R_2	• • •	R_n
U_1	×			
U_2	×			
U_3		×		×
U_4				×
U_5				×
U_6				×
⋮				
U_m	×			

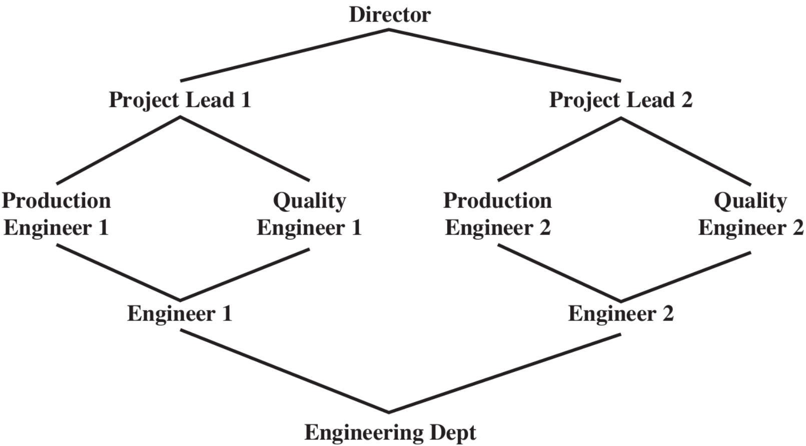
OBJECTS

ROLES	R_1	R_2	R_n	F_1	F_1	P_1	P_2	D_1	D_2
R_1	control	owner	owner control	read *	read owner	wakeup	wakeup	seek	owner
R_2		control		write *	execute			owner	seek *
⋮									
R_n			control		write	stop			

Credit: Figure 4.8 in Stallings and Brown, *Computer Security*, 2nd Ed., Pearson 2012

Hierarchies in RBAC

- Hierarchy of an organisation can be reflected in roles
- A higher role includes all access rights of lower role



Credit: Figure 4.10 in Stallings and Brown, *Computer Security*, 2nd Ed., Pearson 2012

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  - maximum number of roles that can be granted particular access rights
- Prerequisite: condition upon which user can be assigned a role, e.g.
  - user can only be assigned a senior role if already assigned a junior role

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Mandatory Access Control

- Based on multilevel security (MLS)

  top secret > secret > confidential > restricted > unclassified

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- Bell-LaPadula **Confidentiality** model and Biba **Integrity** model

Implementations of MAC

- SELinux: Linux kernel modules available to most Linux distributions (RedHat, Debian, Ubuntu, SuSE, ...)
- AppArmor: some Linux distributions (Ubuntu, SuSE)
- TrustedBSD: FreeBSD, OpenBSD, OSX, ...
- Mandatory Integrity Control: Vista, Windows 7, Windows 8

Contents

Key Points

- Access control to prevent unauthorized use of resources (objects) by subjects
- Subjects are processes on behalf of users and applications
- Classes of subjects: owner, group, world
- Objects: files, database records, disk blocks, memory segments, processes, . . .
- Access rights: read, write, execute, delete, create, . . .
- DAC: access rights may be granted to other subjects (common in operating systems and databases)
- RBAC: subjects take on role; access rights assigned to roles
- MAC: subjects/objects assigned to levels; subjects cannot modify assignment (e.g. military classification)

Security Issues

- Rely on correct assignment of capabilities/levels to subjects and objects by human administrator

Areas To Explore

- Trusted Computing and Trusted Platform Module (TPM)
- Secure Boot